



Practical Business Case

Deloitte Spain
Advanced Analytics

Assessment overview

At Deloitte's Advanced Analytics team, we seek candidates with a combination of **analytical reasoning and technical skills**. These two qualities are critical to the success of our team, as they enable individuals to tackle complex problems with both technical precision and strategic insight.

As such, during the assessment, you will be asked to solve a business case where you will **be evaluated on both your analytical reasoning and your technical skills**. We place equal importance on these two competencies, and your performance in each area will be closely assessed.

Here are several **recommendations** to help you excel in each area of the assessment:

- **Analytical reasoning:**
 - Clearly articulate the **rationale behind your decisions** and the approach you've taken.
 - Don't just share numeric results or figures, explain their **meaning, insights and implications**.
 - Be business-driven and reflect on how your findings will **impact the proposed business context**.
- **Technical skills:**
 - Write reproducible code that is **well-structured**, clean, and easy to follow.
 - **Leverage graphs and visualizations** to deepen your comprehension of the problem and to communicate insights more effectively.
 - Instead of making multiple models, invest more time on **choosing one that is fit for purpose** and makes sense.

As a **general guideline**, prioritize **quality over quantity**.

Also, you **may use Gen AI as a support tool**, but ensure that your final outcome meets the standards expected for a Deloitte client deliverable.

Business Case

Imagine you are a Deloitte Consultant working on a project for a **major international telecom company**. The company **provides various mobile phone plans** to a diverse customer base across the US. However, they have recently noticed that a **significant number of customers are leaving to their competitors**, causing revenue losses and an erosion of their market share. They want to take **commercial actions** to retain their high-value customers likely lost to churn, **but they don't know who to contact first**.

Your task in this assessment is to **help them overcome their churn problem** by...

- Understanding the **key characteristics of customers** who are likely to churn.
- Predicting the **customer's likelihood of leaving** (churning).
- Identifying the **top 500 customers to contact first** – maximizing profitability and minimizing churn.
- Evaluating how your strategy compares to **randomly contacting the business' customers**.

You must **answer these business questions in a section at the end of your code by leveraging your data science skills and using the available customer data**. Please, read the next sections carefully as they contain more details about the technical steps required, the delivery format and the dataset.

Technical challenge

To answer the questions, you are expected to use their data to perform **data preparation, modeling, and evaluation** to develop a **predictive model to estimate the churn rate** of their customers.

You must use **Python or R** (via Jupyter Notebook or RMarkdown). We highly value your reasoning skills, and not only your coding abilities. Therefore, include **concise comments in your code to justify your decisions**, explain the **reasoning** behind each step, and **highlight key conclusions**, as all these will **be highly valued by the client**.

Please note that all comments, explanations, and answers must be **written in English** and tailored to your audience. Your insights and conclusions should be accessible to the non-technical company stakeholders.

Beware of the time available, this assessment is time-constrained and completing all the steps can be challenging. Hence, using **efficient and pragmatic** solutions is highly recommended.

Deliverable

After completing the tasks described above, please submit your code with the appropriate explanations in a compressed file via email. Set your name as the files name: *CandidatesName_Candidates_Surname.zip*.

Make sure at **the end of your code there is a section answering the four questions** in a way that is accessible to non-technical stakeholders. If you are using R, submit it as an **R-Markdown** file; if you are using Python, submit it as a **Jupyter Notebook**.

Separately from the compressed file, you also need to attach the Jupyter Notebook or RMarkdown in **HTML** format with your name as the files name.

Data sources

This assessment uses the **churn_all.csv** dataset, which you should have already received. The column names should be self-explanatory. However, if more detail was to be needed, you can find a data table below with the variable names and descriptions.

Variable dictionary

Variable Name	Description
state	Two-letter code from the given USA state
area_code	Telephone area code (prefix)
phone_number	Telephone number
international_plan	The customer has an international plan
voice_mail_plan	The customer has a contracted voice mail service
number_vmail_messages	Number of messages in voice mail
total_day_minutes	Morning – Consumption in minutes
total_day_calls	Morning – Number of calls
total_day_charge	Morning – Amount to pay
total_eve_minutes	Evening – Consumption in minutes
total_eve_calls	Evening – Number of calls
total_eve_charge	Evening – Amount to pay
total_night_minutes	Night – Consumption in minutes
total_night_calls	Night – Number of calls
total_night_charge	Night – Amount to pay
total_intl_minutes	International - Consumption in minutes
total_intl_calls	International - Number of calls
total_intl_charge	International – Amount to pay
number_customer_service_calls	Number of customer service calls